PUBLIC TECHNICAL & INDUSTRIAL COMPANION | VERSION 1.5

Evidence-State Runtime

for Computational Engineering

Recoverable execution, scientific memory, replay assurance and detachable AI on workstation-scale hardware

Joseph Maxwell

The Systemica Institute, London, United Kingdom

ORCID: 0009-0002-4070-6035 | August 2026

Core proposition. Computational evidence should be treated as a stateful object. A trustworthy workflow records what entered, what executed, what completed, what resumed, what was separately identified as replay, what failed, what is retained as scientific evidence, and what downstream action that evidence has earned.

Industrial question. Can an organisation prove what happened, recover interrupted work without changing scientific identity, reproduce the relevant decision, contain unsupported output, and show who authorised the next action?

This companion in plain language. **Question:** how can a long computational workflow show what actually completed, what safely resumed, what repeated, and what action was authorised? **What was exercised:** atomic scientific units, interruption recovery, multi-contract replay, separated memory, and human-controlled evidence updates. **Why it matters:** a successful process exit is not the same as accepted scientific evidence or engineering permission. **Boundary:** this is local implementation and exercise, not enterprise qualification or a public software distribution.

Three ways to read this companion. Orientation: read the callouts, the Spatial example, and the Conclusion. **Operational detail:** read atomic execution, replay, memory, and the furnace example. **Audit depth:** follow the public receipts and programme registers named in the research compendium.

Public implementation boundary. This companion describes demonstrated control logic and operational evidence without publishing the complete private runtime, policy engines, orchestration logic, storage internals, or commercial integration code. Materials is the demonstrated scientific carrier domain. Cross-domain scientific usefulness remains a transfer hypothesis.

Current maturity boundary. The runtime has been implemented and exercised locally on the reported scientific and engineering workflows. It has not yet undergone an external customer pilot, production qualification, security assessment, enterprise deployment, regulated validation, or controlled economic comparison.

Distribution boundary. This document describes a public technical surface of a private implementation; it is not itself a public software distribution.

Why move beyond materials ML? Materials ML provides the deepest scientific evidence because its sources, models, controls, metrics, and replay states are inspectable. The engineering cases test a different question: whether the same control principles remain useful when a result changes geometry, topology, fault response, or the next physical test. The first asks whether an interpretation is justified. The second asks what a result is allowed to change next.

1 Scientific method and runtime assurance are different

A scientific method states what is measured: the source data, cohort, representation, model, split, controls, metrics, and decision rule. Runtime assurance records execution identity, completed units, safe resume, separately identified replay, and the passed comparison contract.

Neither substitutes for the other. A perfectly logged poor method remains poor science. A sound method with missing, duplicated, or ambiguous outputs remains incomplete evidence.

	Scientific method	Runtime assurance
Primary question	What is being tested, against what comparator, and under which assumptions?	Did the frozen test execute completely and reproduce under the declared contract?
Typical objects	Dataset, cohort, features, model, hyperparameters, folds, metrics, nulls, thresholds	Execution identity, unit plan, lock, journal, output contract, hashes, terminal seal, replay comparison
Failure example	Leakage, unsuitable target, weak comparator, post-hoc threshold	Partial output, duplicate unit, identity drift, corrupt checkpoint, replay mismatch
Authority	Supports the scientific interpretation only if its design is adequate	Supports only the declared completion/recovery/replay state

Authority rule. A weaker evidence state does not inherit the authority of a stronger one merely because both occur in the same workflow.

2 Worked public example: exact replay preserves a refusal

The final 256-material spatial confirmation tested two predeclared representation questions with Ridge models. For bulk modulus, continuous metric geometry had to exceed both a baseline and a matched sham by at least 0.05 mean R^2 in each of two population views. For shear modulus, polarity information had to exceed geometry and its sham by at least 0.02 in each view.

The fresh execution committed 576/576 scientific units. A separately identified replay committed another 576/576. All 1,728 corresponding output pairs were byte-identical. Semantic identity passed, and 927,361 numeric cells matched exactly.

Question	Continuity view	Chemical-system view	Frozen decision	Practical reading
Bulk geometry	+0.4778 vs baseline; +0.5145 vs sham	+0.4855 vs baseline; +0.5157 vs sham	PASS both views	Compact metric geometry earned retention for this bounded route.
Shear polarity over geometry	+0.0123; below +0.02	+0.0453; above +0.02	Conditional; cross-view FAIL	One view could not repair failure in the other.

Why this matters industrially. Exact replay proved that the mixed decision itself repeated. It did not grant permission to turn the failed cross-view polarity claim into a universal success. This is the practical purpose of separating execution quality from scientific outcome.

3 What an organisation can inspect

Operational question	Inspectable answer
What ran?	Frozen plan plus source, configuration, dependency, and unit identities.
What completed?	Atomic output checks and a terminal evidence state.
What survived interruption?	Verified same-identity continuation, with incomplete units rejected and rerun.
What was separately identified as replay?	A new replay identity with plan, byte, semantic, and numerical comparison states.
What failed or was held?	Preserved adverse evidence, staging, and bounded successor lineage.
What may happen next?	Explicit claim boundary and separately authorised action.

What this is not. It is not a replacement for a domain solver, workflow manager, experiment tracker, or quality-management system. It is not an autonomous decision maker. It does not establish regulatory compliance, cyber-security certification, production qualification, enterprise scale, or economic return.

4 Demonstrated capability surface

Functional layer	Demonstrated capability	Evidence maturity	Public evidence boundary
Scientific execution	Source-specific materials runners and frozen analysis routes	Local scientific exercise	Real execution; replay strength varies by route
Recovery	Atomic completion, verified resume, terminal verification	Local implementation and exercise	Demonstrated in bounded executions
Evidence representation	Immutable evidence references and explicit claim boundaries	Local implementation	Public semantics; private implementation omitted
Candidate outcomes	Retain, exclude, hold, and request-evidence states	Local implementation and exercise	Decision meanings public; engine private
Action authority	Explicit bounded approval or refusal	Local engineering exercise	Human-authorised next action only
Scientific memory	Separately authorised retention of accepted evidence references	Local implementation	Concept/state public; storage internals private
Worker isolation	Scoped workspaces and restricted writes	Local implementation	Boundary public; orchestration internals private
State tracking	Typed transition and trajectory controls	Local engineering exercise	Capability evidenced; full implementation not released

5 Evidence-state control model

SOURCE MEMORY ↓ NATIVE DOMAIN EXECUTION ↓ EXECUTION MEMORY ↓ CANDIDATE EVIDENCE ↓ REPLAY / ADJUDICATION ↓ HELD OR ACCEPTED EVIDENCE ↓ SEPARATELY AUTHORISED ACTION

The domain supplies meanings, admission rules, models, and application thresholds. The portable layer preserves identity, completion, comparison, lineage, and action state around those declared objects.

State	Entry requirement	Maximum justified use
Available	A source object exists	Discovery only
Admitted	Provenance, units, methods, and membership satisfy domain rules	Enter the declared analysis
Executed	A unit produced staging output	No completion claim
Committed	Identity, schema, uniqueness, and expected output pass	Measured local evidence
Replay-adjudicated	New identity executes and declared comparisons are evaluated	Only the declared replay state
Held	A required gate fails or remains unresolved	Retain; do not advance
Action-bounded	Human authorisation satisfies scope and evidence boundary	Only the declared next action

6 Atomic execution and recovery

The reference design binds source, configuration, dependencies, ordered units, and expected outputs into an execution identity. A unit advances only after its complete output set passes verification. Incomplete staging is never reused as completed evidence.

Control	Public behaviour	Industrial consequence
Single writer	One process owns a declared execution	Concurrent processes cannot silently own the same run
Atomic completion	Complete outputs accepted only after validation	Partial files cannot impersonate completion
Hash-linked journal	Checkpoint events bind continuation to prior state	Continuation lineage remains inspectable
Terminal verification	Outputs and receipts rechecked before sealing	Post-run inconsistency remains detectable
Resume	Same unsealed identity; verified units skipped; incomplete units rerun	Interruption does not create a new scientific story
Replay	New identity and predecessor; no predecessor output reuse	Repeatability becomes separately inspectable
Duplicate/output refusal	Missing, duplicate, unexpected, or schema-invalid output fails	Restart cannot silently double-count or accept extras

Operational distinction. Resume proves recoverability. Replay tests predecessor-linked repeatability. A resumed run must never be described as a separately identified repeat or external replication.

7 Multi-contract replay

Reproducibility is not one Boolean flag. A route can be scientifically numerically equivalent while its files differ in serialization, or byte-identical while reproducing a failed scientific threshold.

Contract	Question	Failure meaning
Plan identity	Did the same complete ordered workload execute?	Subset, superset, or reordered route
Source/configuration	Were frozen inputs and settings preserved?	Different scientific identity
Schema/coverage	Is every expected object present exactly once?	Incomplete or ambiguous output
Semantic identity	Do values refer to corresponding scientific cells?	Scientifically misaligned comparison
Numerical equivalence	Do numeric cells satisfy the frozen rule?	Numerical replay contract failed
Raw-byte identity	Are payloads byte-identical?	Storage divergence; not automatically scientific divergence

8 Memory separated by authority

Memory class	Contains	What it may define
Source memory	Dataset identity, cohort, provenance, units, methods	What entered
Execution memory	Plans, checkpoints, receipts, hashes, failures, predecessor	What executed
Scientific evidence memory	Outputs, controls, replay states, holds, claim boundaries	Accepted scientific evidence
Governed persistent memory	References to accepted evidence under separate authorisation	Never more than the source evidence supports
Narrative/working memory	Summaries, discussion, hypotheses, prose, AI interpretation	Orientation only

Governing rule. Narrative memory may help reconstruct the work. It cannot overwrite scientific evidence memory.

9 Bounded authority and detachable generative AI

The negative route is deliberately simple: unsupported request → HOLD → zero accepted-memory writes → no downstream action. The positive route requires verified evidence, explicit human authorisation, bounded persistence, read-back verification, and preserved reversible history.

AI-assisted development is not AI-dependent scientific execution. Conventional fitted models are part of the science. Generative systems may orient, critique, diagnose, or propose, but receive no permission to promote their own output into accepted evidence or engineering action.

Role	Examples	Authority
Deterministic control	Identity, validation, receipts, replay rules	Declared rules only
Conventional fitted ML	Ridge, tree models, MLPs	Scientific output under frozen protocol
Optional generative assistance	Orientation, critique, diagnosis, proposals	Advisory; detachable
Human authority	Protocol freeze, interpretation, engineering approval	Final bounded decision

10 Empirical assurance and compute record

Route	Execution/comparison	Observed state
JARVIS-DFT	48 fresh + 48 replay; 28,248 cells; 192 files	Maximum delta about 1.07×10^{-14}; 60 files byte-divergent; numerical equivalence retained
Materials Project	64 fresh + 64 replay; 37,924 cells	916 strict exceedances, all elastic; mixed replay retained
Matbench V0.3	3,440 fresh + 3,440 repeat; 65,740 cells	Zero numerical differences; semantic identity pass
Spatial V0.5	256 fresh + 256 replay	Byte, semantic, and numerical identity pass
Spatial V0.6.1	576 fresh + 576 replay; 1,728 pairs; 927,361 cells	Exact replay; bulk geometry passes both views; shear polarity conditional

A scientific unit is an evidence boundary, not a fixed amount of computation.

Route	Measured execution	Interpretation boundary
JARVIS	About 9.93 h; eight bounded CPU workers	Workstation audit, not provider DFT
Matbench	19.16 h fresh; 20.60 h repeat; two CPU workers	Complete removal workload; no GPU dependency
Spatial V0.5	9.64 min fresh; 9.10 min replay	Smaller factorial study
Spatial V0.6.1	20.67 h fresh; 16.20 h replay committed-unit spans	Not launcher-to-terminal wall times

Principal workstation: AMD Ryzen 7 5800X, 32 GB RAM, NVIDIA RTX 3060 installed. The reported tabular and Matbench routes were CPU-only where receipts support that statement. Published first-principles-derived records were consumed; provider DFT was not recomputed locally.

11 Beyond Materials ML: Engineering Execution Witnesses

Three earlier studies provide local engineering exercises of case identity, invalid-transition refusal, failure retention, and bounded next-action logic. They did not retrospectively execute the complete present-day implementation.

Case	Consequential correction or finding	Bounded next action
Furnace viewport	Correcting 350–375 mm to 35–40 mm invalidated transfer of an inherited 59.2°C result; active cooling added a failure dependency	HOLD; test an opaque guarded thermal-interface coupon
Dual tank	Fresh reserve, feeder/recovery, and returned/consumed states had to remain provenance-distinct	HOLD; low-pressure non-potable tracer rig
Heat shield	Segmentation traded thermal performance for fault isolation rather than improving both	HOLD; matched cold-flow/heated coupons including blocked-cell and contained-leak cases

Across the cases, 13 candidate objects, five descendants, 116 scenario runs, 9,150 parameter/uncertainty runs, and at least 9,380 model operations narrowed broad concepts to falsifiable tests. These counts show bounded computation, not labour, cost, or calendar savings.

12 Worked public example: a bounded furnace workflow

The furnace viewport study shows how a completed calculation can remain ineligible for fabrication. This is public contract-level semantics, not private orchestration.

1. **Declare the case.** Corrected interface geometry: 35–40 mm. Source assumptions and thermal boundaries are declared. Requested decision: is the concept mature enough to fabricate or test?
2. **Bind the computation.** A deterministic thermal model, candidate families, uncertainty/failure scenarios, and expected outputs are declared.
3. **Run and retain failure.** Baseline response, contact/seal/emissivity uncertainty, active-cooling-loss, and edge/frame-transfer uncertainty remain visible.
4. **Compare predecessor evidence.** The inherited 59.2 °C result belonged to a 350–375 mm geometry. The corrected geometry creates a different scientific identity, so the prior result is retained historically but cannot transfer silently.
5. **Adjudicate.** The model may be numerically complete while interface and cooling dependencies remain unresolved. Outcome: HOLD.
6. **Authorise only the bounded action.** A guarded opaque thermal-interface coupon is authorised; full fabrication, furnace deployment, and qualification are not.

```
case_id: furnace_viewport_corrected_geometry
source_state: declared
geometry_state: corrected_35_40_mm
execution_state: complete
uncertainty_state: unresolved_edge_and_cooling_dependency
prior_result_transfer: refused
evidence_state: HOLD
authorised_next_action: guarded_coupon_test
```

Boundary. This pseudo-record is illustrative public semantics, not the private production schema or orchestration implementation.

12.1 How to translate this to another engineering workflow

Ask five questions: What exact object is being evaluated? What source or configuration makes this run scientifically distinct? Which failures must remain visible rather than averaged away? Which predecessor evidence may transfer, and what changed enough to break transfer? What is the smallest next action the evidence justifies?

Furnace example	Generic engineering translation
Corrected interface geometry	Changed design/configuration identity
Thermal solver	Native domain solver
Cooling-loss scenario	Declared failure or perturbation case
Inherited 59.2 °C result	Predecessor evidence
Transfer refused	Evidence identity mismatch
Coupon test	Bounded next action

Dual-tank micro-example. A numerically successful flow route can still be refused if the state transition itself is invalid.

13 Scientific units, fits, and accepted boundaries

Scientific unit. The smallest contract-bound part of a declared experiment that may complete independently. It binds scientific coordinates, required outputs, and execution identity; it commits only after output, schema, identity, and atomic-commit checks pass.

A unit is not a fixed quantity of computation. Matbench V0.2 makes the distinction concrete: 20 official-fold units each contained one baseline fit and five target-permutation refits, so **20 units represented 120 estimator fits**. Matbench V0.3 contained 20 baseline/perturbation, 2,840 individual-removal, and 580 group-removal units per identity. Spatial V0.6.1 contained 144 valid and 432 paired-null units per identity.

DECLARED UNIT → LOCAL STAGING → FILE/SCHEMA/IDENTITY/HASH CHECKS → ATOMIC COMMIT → JOURNAL/CHECKPOINT

Invalid or incomplete staging is retained as failure evidence and recomputed; it does not advance the checkpoint. **Resume from the last verified scientific boundary, not from the last visible file.**

14 Programme scale without a synthetic mega-number

Route	Fresh	Repeat/replay	Unit executions	Other evidenced work
JARVIS	48	48	96	28,248 numeric cells; 192 file pairs
Materials Project	64	64	128	37,924 cells; 916 elastic exceedances
Matbench V0.2	20	0	20	120 estimator fits
Matbench V0.3	3,440	3,440	6,880	6,880 fits; 65,740 cells
Spatial V0.5	256	256	512	exact declared replay
Spatial V0.6.1	576	576	1,152	1,152 fits; 927,361 cells
Other spatial predecessors	28	20	48	V0.1–V0.4 retained separately
Total	4,432	4,404	8,836	Unlike objects remain separate

Of the 4,404 later executions, 3,440 are same-environment repeat units and 964 are predecessor-linked replay units. At least 1,059,273 explicitly counted numeric fresh-to-repeat/replay cells were compared. Neither total is a model-operation count. Engineering evidence is also separate: 13 candidates, five descendants, 116 scenarios, 9,150 parameter/uncertainty runs, and at least 9,380 model operations.

15 Observed continuity on a shared workstation

The JARVIS replay was interrupted and resumed under the same identity; incomplete staging was rejected and recomputed. The Materials Project replay similarly retained only verified complete units across a later resume launch. A controlled-fixture matrix terminated child processes at 11 declared fault boundaries and recovered each fixture to three unique units. The latter is controller fault-injection evidence, not 11 scientific experiments.

The principal workstation was shared rather than dedicated. Calendar time included ordinary use, review, interruption, and unknown gaps. It must not be reported as active runtime. Elapsed process time, summed unit wall time, committed-unit span, and calendar elapsed are different quantities. Spatial V0.6.1's 20.67 h fresh and 16.20 h replay values are committed-unit spans.

`[RUN] - [ordinary use/review] - [RUN] - [interrupt/checkpoint] - [RESUME] - [analysis] - [REPLAY]`

Boundary. Resume tests continuation of one identity. Repeat creates a separately identified same-environment execution where applicable. Replay links a new identity to a predecessor under declared comparison contracts. Resume is not replication, and replay is not automatically external replication.

16 Worked human-authority example: replay is not self-promotion

A chronologically separate post-replay transaction exercised the private implementation without changing historical run receipts.

EXECUTION → COMPARISON → CANDIDATE CLAIM → HUMAN DECISION → BOUNDED WRITE OR HOLD → READ-BACK

Test	Input	Observed control	Result
No human confirmation	Eligible Matbench claim	Promotion gate refused before persistence	zero writes
Explicit founder decision	Matbench and Spatial bulk bounded claims	Exact claim and ceiling persisted, predecessor retained, read-back matched	two bounded writes; no engineering action
Ceiling violation	Spatial shear and MP elasticity strengthened beyond evidence	Claim-ceiling gate refused both attempts	zero positive promotions

The accepted Matbench claim is limited to weak rank alignment under the frozen protocol and exact same-environment repeat. The accepted Spatial claim is limited to bulk metric geometry under two required views and exact local replay. Spatial shear remains conditional; Materials Project elasticity remains held. Execution completion may update execution memory, but only explicit human adjudication can update accepted scientific memory. Even then, engineering permission remains separate.

17 Engineering models behind the witnesses

Case	Physical object	Native model	Failure exposed	Next test
Furnace	High-temperature viewport/interface	Dominant-path thermal screening; six scenarios; 9,000 uncertainty samples; 9,120 operations	Geometry invalidated inherited result; cooling dependency; unresolved edge/frame path	guarded opaque coupon
Dual tank	Reserve/feeder/recovery topology	Lumped flow/state model; typed transitions; 26 scenarios; 70 parameter cases	Recovered water collapsed into reserve; invalid return paths	low-pressure tracer rig
Heat shield	Segmented transpiration coupon	Two-node transient thermal cells; Darcy-style flow; fault topology; 84 scenarios; 80 cases	Nominal cooling and fault localisation diverged	matched blocked/leak coupons

The furnace route varied geometry, aperture, emissivity, contact and seal assumptions, hot-side boundary, cooling loss, and edge/frame transfer. The tank route separated source provenance, residence, return, gate, consumed, and contained states; its descendant improved an exploratory worst service margin from -0.266 to -0.090 without resolving final sizing. The heat-shield monolith produced a 175.3–389.5 K nominal reduction range, whereas the segmented descendant included -83.5 to 245.4 K while improving local fault information. These are bounded screening instruments, not physical validation.

The technical contribution is the programme's design synthesis and engineering reduction, not a claim of literature-level novelty for every physical mechanism.

18 Practical industrial use patterns

Setting	Operational failure	Pilot control pattern	Decision artefact
R&D screening	Attractive score advances expensive work too early	Admission, viability, replay, action gates	Retain/hold/request evidence
Overnight compute	Interrupted outputs merged manually	Atomic completion and verified resume	Recovery receipt
Restricted data	Data cannot leave controlled environment	Local native science; optional AI detached	Offline manifest
Multi-tool engineering	Tools disagree about completion	Output contracts and terminal checks	Completeness receipt
AI-assisted teams	Generated summary outruns evidence	Narrative/evidence separation	Claim-boundary record
Safety/cost decisions	Repeat mistaken for permission	Replay separate from human approval	Action authorisation
Incident review	Input-to-decision chain cannot be reconstructed	Source, execution, evidence lineage	Audit pack

19 A bounded industrial pilot

The defensible first pilot wraps one existing computational or AI-assisted technical workflow. It does not replace the organisation's solver or data system.

Step	Activity	Deliverable
1	Map sources, tools, human roles, and current permission points	Workflow and authority map
2	Freeze one plan and output contract	Execution identity and baseline manifest
3	Inject interruption, missing output, and unsupported action	Failure/containment evidence
4	Resume from verified checkpoints	Recovery receipt and duplicate check
5	Run a separately identified replay	Byte/semantic/numerical comparison
6	Exercise HOLD and one explicitly authorised action	Zero-write and positive-route receipts
7	Measure time, rework, false advancement, and audit effort	Pilot evidence boundary

Pilot success criterion. Demonstrate bounded controls on one declared workflow and measure overhead against its existing practice. Do not claim universal safety, enterprise ROI, or regulatory compliance.

20 Integration, trust, and observability boundaries

The intended integration model does not require an existing workflow to adopt Systemica scientific models. A bounded integration maps the workflow into source/configuration identity, expected outputs, comparison rules, and action boundaries while leaving domain computation native.

Risk/need	Present public-facing control	Still required for production
Corrupt payload	Schema, contract, hash verification	Storage hardening and monitored integrity
Concurrent writer	Single-writer ownership	Host/process identity policy
Dependency drift	Dependency identity	SBOM, signed builds, vulnerability response
Credential exposure	Local bounded operation	Secret policy, rotation, audit
Malicious worker	Scoped work, no self-promotion	Sandbox hardening and least privilege
Audit tampering	Hash-linked receipts, immutable packs	External signature or WORM anchoring
Progress/capacity	Unit/checkpoint telemetry, bounded workers	Common telemetry, memory-aware cross-run scheduling
Scale	Local CPU execution	Common GPU, cluster, cloud, distributed service

Evidence integrity is not cyber-security. Current evidence does not establish penetration-tested security, air-gap qualification, export-control compliance, enterprise access management, or regulatory certification.

21 Measuring industrial value

The controls are demonstrated; economic value is not yet measured. A serious pilot establishes a comparator before claiming savings.

Measure	Operational definition	Why it matters
Time to recover	Interruption to next verified committed unit	Resilience
Rework avoided	Verified units safely skipped	Retained compute/labour
False advancements blocked	Unsupported outputs prevented from accepted state	Decision-risk control
Audit reconstruction time	Time to rebuild input-plan-output-action chain	Assurance overhead
Escalation precision	Cases sent to higher fidelity versus held/rejected	Compute/test allocation
Decision latency	Evidence arrival to authorised next action	Throughput
Operator burden	Review minutes and unresolved exceptions	Prevents hidden automation overhead

No claim of ROI, labour saved, production throughput, or reduced engineering error is supported until these measures are collected against a conventional comparator.

22 Adoption path and open gates

Stage	Minimum capability	Value added
Observe	Stable IDs and source/config capture	Know who ran what
Contract	Expected outputs and atomic completion	Partial files stop impersonating completion
Recover	Journal and verified resume	Continuation without identity ambiguity
Replay	New identity and multi-contract comparison	Repeatability becomes inspectable
Separate memory	Source, execution, evidence, narrative classes	Interpretation cannot overwrite science
Gate action	Evidence boundary and human approval	Computation stops becoming permission
Optional AI	Bounded proposal/diagnosis interface	Acceleration without control of scientific state

Open gates include cross-platform replay, a common recovery adapter across runners, matched comparison with established workflow systems, a prospective customer pilot, a second scientific domain, security qualification, tensor/method-resolved Materials Project elasticity, common continuous telemetry, and measured operational value.

23 Conclusion

The Evidence-State Runtime is an assurance architecture around scientific computation, not another scheduler and not yet a universal production platform. It makes identity, completion, recovery, replay, memory, failure, and action permission separately inspectable.

Its practical industrial proposition is incremental. Existing scientific and AI-assisted workflows can be wrapped so that interruption, repeatability, retained evidence, and human permission cease to be implicit. The system does not make every computation trustworthy. It makes the state of a computation clear enough that trust and escalation need not be inferred from successful termination alone.

> **Bottom line.** Preserve the work. Preserve the failure. Preserve the distinction between evidence and permission.

If you remember one thing: the runtime should make the earned evidence state visible without allowing execution, replay, AI assistance, or narrative confidence to authorise a stronger action automatically.

References

1. Koster, J. and Rahmann, S. Snakemake—a scalable bioinformatics workflow engine. *Bioinformatics* 28, 2520–2522 (2012).
2. Di Tommaso, P. et al. Nextflow enables reproducible computational workflows. *Nature Biotechnology* 35, 316–319 (2017).
3. Amstutz, P. et al. Common Workflow Language, v1.0. Specification (2016).
4. Lebo, T. et al. PROV-O: The PROV Ontology. W3C Recommendation (2013).
5. Soiland-Reyes, S. et al. Packaging research artefacts with RO-Crate. *Data Science* 5, 97–138 (2022).
6. Wilkinson, M. D. et al. The FAIR Guiding Principles. *Scientific Data* 3, 160018 (2016).
7. Tabassi, E. Artificial Intelligence Risk Management Framework (AI RMF 1.0). NIST AI 100-1 (2023).
8. Peng, R. D. Reproducible research in computational science. *Science* 334, 1226–1227 (2011).

Declarations and evidence boundary

Funding. This research was conducted independently without dedicated external research grant funding.

Competing interests. The author is founder of a venture developing software and services related to scientific evidence assurance and engineering decision support described in this work.

AI assistance. AI and coding tools assisted code development, critique, diagnosis, evidence organisation, and manuscript preparation. They did not generate source target data. Frozen analyses were executed by conventional scientific software under declared protocols, and scientific interpretations and conclusions were reviewed by the author. Optional generative systems may orient, critique, diagnose, or propose, but they are not permitted to promote their own outputs into accepted scientific evidence or engineering action.

Author contribution. Joseph Maxwell is the sole human author and research lead of the reported programme.

Public disclosure boundary. This companion intentionally omits the complete private runtime, proprietary orchestration, internal persistence transactions, general policy engines, and reusable commercial integration source. It does not authorise deployment, scientific promotion, regulatory claims, or customer-performance claims.